

DISCRETE MATHEMATICS

10 Important Two-Mark Questions & 4 Important Sixteen-Mark Questions — Each Unit

Regulation 2025 • Exam-Oriented Question Bank

Note: These are high-priority revision questions based on the topics covered in the prepared Regulation-2025 notes. They are not a guarantee of the exact university question paper.

UNIT I — MATHEMATICAL LOGIC

A. TEN IMPORTANT 2-MARK QUESTIONS

1. Define proposition, simple proposition and compound proposition.
2. Write the truth table for $p \rightarrow q$.
3. Define tautology, contradiction and contingency with examples.
4. State De Morgan's laws in propositional logic.
5. Define logical equivalence. How is it verified using a truth table?
6. What are predicates and quantifiers? Define $\forall$ and $\exists$.
7. Write the negation rules for universal and existential quantifiers.
8. State Modus Ponens and Modus Tollens.
9. Differentiate converse, inverse and contrapositive of $p \rightarrow q$.
10. What is mathematical induction? State its two essential steps.

B. FOUR IMPORTANT 16-MARK QUESTIONS

1. Construct the truth table and prove a given compound proposition is a tautology.
2. Prove important logical equivalences using truth tables and laws of logic.
3. Convert a propositional expression into CNF and DNF (and principal forms where required).
4. Translate English statements into predicate logic and prove validity using quantifiers and rules of inference.

UNIT II — SETS

A. TEN IMPORTANT 2-MARK QUESTIONS

1. Define a set, subset, proper subset and power set.
2. Find the power set of a small finite set.
3. Define union, intersection, difference and complement of sets.
4. State De Morgan's laws for sets.
5. Define Cartesian product and give its cardinality formula.
6. State the formula for $|A \cup B|$.
7. State the inclusion-exclusion formula for three finite sets.
8. Distinguish finite, infinite, empty and universal sets.
9. Prove or verify a basic set identity using set laws.
10. If $|A| = n$, find $|P(A)|$.

B. FOUR IMPORTANT 16-MARK QUESTIONS

1. Prove important set identities using algebra of sets and De Morgan's laws.
2. Solve a three-set cardinality problem using the inclusion-exclusion principle.
3. Find and represent Cartesian products and power sets for given finite sets, including cardinalities.
4. Solve a Venn-diagram based counting problem involving union, intersection, complements and differences.

UNIT III — RELATIONS & FUNCTIONS

A. TEN IMPORTANT 2-MARK QUESTIONS

1. Define a binary relation from A to B.
2. Define reflexive, symmetric, antisymmetric and transitive relations.
3. What is an equivalence relation?
4. What is a partial order relation? Give its properties.
5. What is an equivalence class?
6. Define reflexive, symmetric and transitive closures.
7. What is a function $f:A \rightarrow B$?
8. Define one-one, onto and bijective functions.
9. When does an inverse function exist?
10. Define composition of functions.

B. FOUR IMPORTANT 16-MARK QUESTIONS

1. Given a relation, test whether it is reflexive, symmetric, antisymmetric and transitive, with justification.
2. Find the transitive closure of a relation using Warshall's algorithm.
3. Prove that a relation is an equivalence relation and determine its equivalence classes.
4. Test a function for injectivity/surjectivity, find its inverse when it exists, and evaluate compositions.

UNIT IV — GRAPH THEORY

A. TEN IMPORTANT 2-MARK QUESTIONS

1. Define graph, vertex, edge and degree of a vertex.
2. State the Handshaking Theorem.
3. Define simple graph, multigraph and pseudograph.
4. Find the number of edges in K_n .
5. Define walk, trail, path and circuit.
6. What is a connected graph?
7. State the conditions for an Euler circuit and Euler trail.
8. Define Hamiltonian path and Hamiltonian cycle.
9. Define adjacency matrix and incidence matrix.
10. Define a tree and state the number of edges in a tree with n vertices.

B. FOUR IMPORTANT 16-MARK QUESTIONS

1. Use the Handshaking Theorem to determine missing degrees/number of edges and prove that the number of odd-degree vertices is even.
2. Determine whether a given graph has an Euler trail/circuit and construct it when possible.
3. Represent a given graph using adjacency and incidence matrices and interpret the matrices.
4. Prove important properties of trees and find/construct a spanning tree of a connected graph.

UNIT V — COMBINATORICS

A. TEN IMPORTANT 2-MARK QUESTIONS

1. State the Rule of Sum and Rule of Product.
2. Define factorial and state $0!$.
3. Write the formula for permutations $P(n,r)$.
4. Write the formula for combinations $C(n,r)$.
5. Distinguish permutation from combination.
6. State the symmetry property of binomial coefficients.
7. State Pascal's identity.
8. State the Binomial Theorem and its general term.
9. State the Pigeonhole Principle.
10. State the Inclusion-Exclusion Principle for two and three sets.

B. FOUR IMPORTANT 16-MARK QUESTIONS

1. Solve counting problems using the fundamental counting principles, permutations and combinations.
2. Solve arrangement problems involving repeated objects and/or restrictions.
3. Expand a binomial and find a specified term or coefficient using the Binomial Theorem.
4. Apply the Pigeonhole and Inclusion-Exclusion principles to solve counting problems.

UNIT VI — RECURRENCE RELATIONS

A. TEN IMPORTANT 2-MARK QUESTIONS

1. Define a recurrence relation and initial conditions.
2. Give an example of a first-order recurrence relation.
3. Distinguish homogeneous and non-homogeneous recurrence relations.
4. Write the characteristic equation for a second-order linear homogeneous recurrence.
5. State the solution form for distinct characteristic roots.
6. State the solution form for repeated characteristic roots.
7. What is a particular solution?
8. What is a complementary (homogeneous) solution?
9. Define the ordinary generating function of a sequence.
10. Why are initial conditions necessary in solving recurrence relations?

B. FOUR IMPORTANT 16-MARK QUESTIONS

1. Solve a first-order linear recurrence relation using the standard method and initial condition.
2. Solve a second-order homogeneous recurrence relation with distinct characteristic roots.
3. Solve a second-order homogeneous recurrence relation with repeated characteristic roots.
4. Solve a non-homogeneous recurrence relation and determine the constants using initial conditions; use generating functions where required.

LAST-MINUTE REVISION PRIORITIES

- Logic: truth tables, equivalence laws, CNF/DNF, quantifiers and rules of inference.
- Sets: laws, De Morgan's laws, power sets, Cartesian products and inclusion-exclusion.
- Relations & Functions: property testing, equivalence/partial order, Warshall's algorithm, inverse and composition.
- Graphs: Handshaking theorem, Euler conditions, Hamilton concepts, matrices and trees.
- Combinatorics: $P(n,r)$, $C(n,r)$, repeated arrangements, Binomial Theorem, Pigeonhole and Inclusion-Exclusion.
- Recurrences: characteristic equation, distinct/repeated roots, non-homogeneous method and generating functions.
- Practice writing 16-mark answers with definition → theorem/formula → working → conclusion.